ASSIGNMENT 9

Advanced SQL Queries Using Joins and Subqueries

HOSPITAL MANAGEMENT SYSTEM

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Course: BCA (Hons)

Subject: Database Management System

1. Objective

Use joins and subqueries to produce meaningful reports across the Hospital Management System. Run these queries after the schema and sample data from Assignments 5 and 7 have been created.

Report 1 — appointment register (multiple joins)

```
SELECT a.appointment_id, p.name AS patient_name,
       d.name AS doctor_name, dep.department_name,
       a.appointment_date, a.appointment_time, a.status FROM
Appointments a
JOIN Patients p ON p.patient_id = a.patient_id
JOIN Doctors d ON d.doctor_id = a.doctor_id
JOIN Departments dep ON dep.department_id = d.department_id
ORDER BY a.appointment_date, a.appointment_time;
```

Explanation / expected output: Combines appointment, patient, doctor and department details into one readable register. Expected: 6 rows.

Report 2 — medicines prescribed at each appointment

```
SELECT a.appointment_id, p.name AS patient_name,
       m.medicine_name, pr.dosage, pr.duration, pr.quantity FROM
Prescriptions pr
JOIN Appointments a ON a.appointment_id = pr.appointment_id
JOIN Patients p ON p.patient_id = a.patient_id
JOIN Medicines m ON m.medicine_id = pr.medicine_id
ORDER BY a.appointment_id, m.medicine_name;
```

Explanation / expected output: Shows each prescription line with patient and medicine details. Expected: 6 rows.

Report 3 — bill summary by patient

```
SELECT p.patient_id, p.name,
       COUNT(b.bill_id) AS bill_count,
       COALESCE(SUM(b.total_amount), 0) AS total_billed
FROM Patients p
LEFT JOIN Appointments a ON a.patient_id = p.patient_id
LEFT JOIN Bills b ON b.appointment_id = a.appointment_id
GROUP BY p.patient_id, p.name
ORDER BY total_billed DESC;
```

Explanation / expected output: LEFT JOIN keeps patients with no bills in the report; COALESCE displays zero instead of NULL. Sample patient totals depend on the appointment-to-bill links.

Report 4 — average feedback rating per doctor

```
SELECT d.doctor_id, d.name,
       COUNT(f.feedback_id) AS feedback_count,
       ROUND(AVG(f.rating), 2) AS average_rating
FROM Doctors d
LEFT JOIN Feedbacks f ON f.doctor_id = d.doctor_id
GROUP BY d.doctor_id, d.name
ORDER BY average_rating DESC;
```

Explanation / expected output: Aggregates feedback by doctor. Doctors with no feedback remain in the report, with NULL average_rating.

Report 5 — correlated subquery: appointments above the average count per doctor

```
SELECT d.doctor_id, d.name
FROM Doctors d
WHERE (SELECT COUNT(*)
       FROM Appointments a
       WHERE a.doctor_id = d.doctor_id) >
      (SELECT AVG(appointment_count)
       FROM (SELECT COUNT(*) AS appointment_count
             FROM Appointments
             GROUP BY doctor_id) AS doctor_counts);
```

Explanation / expected output: Returns doctors whose appointment count is above the average count among doctors who have appointments. With sample data, doctor 2 has 3 appointments and qualifies.

Report 6 — subquery: medicines never prescribed

```
SELECT m.medicine_id, m.medicine_name
FROM Medicines m
WHERE NOT EXISTS (
  SELECT 1 FROM Prescriptions pr
  WHERE pr.medicine_id = m.medicine_id
)
ORDER BY m.medicine_id;
```

Explanation / expected output: Identifies medicines with no matching prescription line. With the sample data, medicines 204 and 205 are not prescribed? Medicine 205 appears in prescription 306, so the expected result is medicine 204 only.

Report 7 — bills above the average bill amount

```
SELECT bill_id, appointment_id, total_amount, payment_status
FROM Bills
WHERE total_amount > (SELECT AVG(total_amount) FROM Bills)
ORDER BY total_amount DESC;
```

Explanation / expected output: Average bill is 580.00, so bills 401 (850.00), 403 (750.00) and 402 (600.00) are returned.

Report 8 — appointments without a bill

```
SELECT a.appointment_id, a.appointment_date, a.status
FROM Appointments a
LEFT JOIN Bills b ON b.appointment_id = a.appointment_id
WHERE b.bill_id IS NULL
ORDER BY a.appointment_id;
```

Explanation / expected output: Finds appointments that have no bill row. In the sample data, appointment 106 has no bill.

2. Query design notes

- INNER JOIN returns only rows with matching records in every joined table. LEFT JOIN preserves every row from theleft-hand table.
- GROUP BY must include the non-aggregated selected columns under MySQL's ONLY_FULL_GROUP_BY mode.
- NOT EXISTS is a robust way to locate medicines without prescription rows.

- The derived table in Report 5 calculates the average number of appointments per doctor who has at least oneappointment.

Conclusion

The reports combine related entities, calculate summary statistics and use subqueries to answer operational questions about appointments, prescriptions, billing, feedback and medicine usage.

ASSIGNMENT 9 — OUTPUTS AND VERIFICATION

Report 1 — Appointment register

ID	Patient	Doctor	Department	Date	Time	Status
101	Aarav Sharma	Dr. Kavya Menon	Cardiology	2026-10-20	09:00	Completed
102	Meera Nair	Dr. Rohan Iyer	General Medicine	2026-10-20	10:30	Completed
103	Kabir Rao	Dr. Kavya Menon	Cardiology	2026-10-21	11:00	Scheduled
104	Ananya Das	Dr. Sana Khan	Orthopaedics	2026-10-22	14:00	Cancelled
105	Ishaan Patel	Dr. Rohan Iyer	General Medicine	2026-10-23	15:30	Scheduled
106	Aarav Sharma	Dr. Rohan Iyer	General Medicine	2026-10-24	09:30	Scheduled

doctor_id	name

2	Dr. Rohan Iyer

Report 2 — Prescriptions by appointment

Appointment	Patient	Medicine	Dosage	Duration	Qty
101	Aarav Sharma	Paracetamol 500 mg	500 mg	3 days	6
101	Aarav Sharma	Omeprazole 20 mg	20 mg	5 days	5
102	Meera Nair	Amoxicillin 250 mg	250 mg	5 days	15
102	Meera Nair	Omeprazole 20 mg	20 mg	7 days	7
103	Kabir Rao	Paracetamol 500 mg	500 mg	2 days	4
104	Ananya Das	Ibuprofen 200 mg	200 mg	3 days	9

Report 3 — Bill summary by patient

Patient ID	Patient	Bill count	Total billed
1	Aarav Sharma	1	850.00
2	Meera Nair	1	600.00
3	Kabir Rao	1	750.00
5	Ishaan Patel	1	500.00
4	Ananya Das	1	200.00

Report 4 — Average feedback rating by doctor

Doctor ID	Doctor	Feedback count	Average rating
1	Dr. Kavya Menon	2	5.00
2	Dr. Rohan Iyer	2	4.00
3	Dr. Sana Khan	1	3.00

Report 5 — Doctors above average appointment count

Report 6 — Medicines never prescribed

medicine_id	medicine_name
204	Calcium Supplement

Report 7 — Bills above average amount (580.00)

bill_id	appointment_id	total_amount	payment_status
401	101	850.00	Paid
403	103	750.00	Pending
402	102	600.00	Partially Paid

Report 8 — Appointments without a bill

appointment_id	appointment_date	status

106	2026-10-24	Scheduled